

Boutiquaat Studio AI Hub

Enterprise v1.0.0

Technical Architecture & Feature Documentation

FRAMEWORK	DATABASE & AUTH	AI ENGINE	LANGUAGE
Next.js 14+ (App Router)	Supabase (PostgreSQL)	RunningHub OpenAPI v2	TypeScript & Tailwind

1. Executive Summary

Boutiquaat Studio AI Hub (`boutiquaat-gen-app`) is a high-performance enterprise AI generation platform engineered for e-commerce, digital advertising, and automated visual production. The platform seamlessly integrates top-tier generative models—spanning text-to-video, image-to-video, background removal, automated social image resizing, and AI creative agents—backed by a resilient backend task queue and real-time state synchronization.

2. Technical Architecture & Stack

Frontend Stack

- **Framework:** Next.js 14+ utilizing App Router architecture with client/server component segregation.
- **Styling & Components:** Tailwind CSS, modern dark glassmorphism styling, and Lucide React icons.
- **State & Monitoring:** Custom React Hooks (`useTasks` , `useAuth`) with automatic task polling and toast notifications.

Backend & Serverless API Routes

- **API Gateway:** Next.js Serverless Route Handlers (`app/api/*`).
- **Database & Storage:** Supabase PostgreSQL (managing `tasks` , `users` , `workflows` , and `image_agent_sessions`) and Supabase Storage buckets.
- **AI API Integrations:**
 - **RunningHub OpenAPI v2:** Execution of AI Apps (`/run/ai-app`), ComfyUI Workflows (`/run/workflow`), Image generation (`rhart-image-*`), and Video generation (`bytedance/*` , `rhart-video/*` , `kling-*` , `google/veo3.1-*`).
 - **RunningHub LLM API:** OpenAI-compatible Chat Completions for prompt enhancement.

3. Key Features & Modules Built

1. Boutiquaat Video AI Studio (`/video`)

A flagship multi-model video generation launcher with automated parameter normalization:

- **Supported Models:**
 - `bytedance/seedance-2.0-global/image-to-video` (Seedance 2.0 Global)

- `seedance-2.0-global-fast/image-to-video` (Seedance 2.0 Fast)
- `rhart-video/sparkvideo-2.0/text-to-video` (SparkVideo 2.0 Text-to-Video)
- `rhart-video/sparkvideo-2.0/image-to-video` (SparkVideo 2.0 Image-to-Video)
- `rhart-video/sparkvideo-2.0/multimodal-video` (SparkVideo 2.0 Multimodal)
- `kling-video-o1/image-to-video` & `kling-v3.0-std-image-to-video`
- `google/veo3.1-pro/start-end-to-video-channel-low-price`
- **Parameter Engine:** Automatic mapping for `resolution` & `quality`, integer parsing for `duration` (4–15s), and removal of empty payload arrays.

2. AI Image Agent Hub (`/image-agent`)

A conversational AI assistant with multi-role specialization:

- **Specialized Roles:** Concept Architect, Lifestyle Brand Manager, Ad Creative Specialist, Anime Illustrator.
- **Capabilities:** Multi-turn image generation, reference image analysis, LLM prompt enhancement, and session persistence in Supabase.

3. Social Resize & PSD Layer Studio (`/studio/social-resize`)

Multi-platform canvas editor supporting automated asset resizing for Instagram Stories (9:16), Feed (1:1), Landscape (16:9), and Banner formats with cross-origin proxy image handling (`/api/proxy-image`).

4. Batch Processing Workflows

- **Batch Image BG Removal:** Parallel ComfyUI background extraction for product catalog imagery.
- **Batch Video BG Removal:** Video green-screen/background stripping yielding transparent MP4 outputs.

5. ComfyUI Workflow Bundling & Auto-Analysis (`/bundling`)

Scans workflow JSON schemas to detect editable node fields, optimize API key channel selection, and launch high-throughput generation tasks.

4. API Reference Summary

Endpoint	Method	Description
<code>/api/runninghub/video-generate</code>	POST	Dispatches video generation tasks to RunningHub Standard API.
<code>/api/runninghub/image-generate</code>	POST	Dispatches image generation tasks (rhart models).
<code>/api/runninghub/upload</code>	POST	Uploads binary media references to RunningHub CDN.
<code>/api/runninghub/query</code>	POST	Fetches task status, execution cost, and output media URLs.
<code>/api/image-agent/sessions</code>	GET/POST	Manages AI Image Agent conversation sessions.
<code>/api/image-agent/enhance</code>	POST	Enhances prompts using LLM chat completions.
<code>/api/tasks</code>	GET/POST	Synchronizes task state with Supabase PostgreSQL database.
<code>/api/proxy-image</code>	GET	CORS proxy for downloading and processing external images.

5. Data Flow Sequence

Task Generation & Execution Lifecycle:

- Media Upload:** Client uploads reference images/videos via `/api/runninghub/upload` to RunningHub CDN.
- Task Initiation:** Client posts payload to `/api/runninghub/video-generate`. Payload is sanitized and sent to `/openapi/v2/{modelPath}`.
- Database Tracking:** Task ID and metadata are registered in Supabase `tasks` table.
- Real-Time Polling:** Frontend `useTasks` hook polls `/api/tasks` until state changes to `SUCCESS` or `FAILED`.

6. Environment Configuration (.env)

```
# RunningHub API Credentials
RUNNINGHUB_BASE_URL=https://www.runninghub.ai
RUNNINGHUB_UPLOAD_URL=https://www.runninghub.cn
RUNNINGHUB_API_KEY_ENTERPRISE=your_enterprise_api_key
RUNNINGHUB_API_KEY_CONSUMER=your_consumer_api_key

# Supabase Infrastructure
NEXT_PUBLIC_SUPABASE_URL=https://your-supabase-id.supabase.co
NEXT_PUBLIC_SUPABASE_ANON_KEY=your_supabase_anon_key
SUPABASE_SERVICE_ROLE_KEY=your_supabase_service_role_key

# Application Settings
NEXT_PUBLIC_APP_URL=http://localhost:3000
```